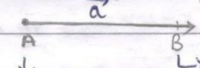


VECTOR ALGEBRA

1+2+2+3+3 = 11 m
definition section formula derivation

Vector : A quantity that has magnitude and direction.

Eg: 
initial point terminal point

$\vec{a}$ has the initial point A moving towards the terminal point B and hence $\vec{a} = \overrightarrow{AB}$ (not $\overrightarrow{BA}$).

$\vec{a}$ in 3D space has 3 components and is written as $\vec{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

Here : Scalar components : a_1, a_2, a_3

Vector components : $a_1\mathbf{i}, a_2\mathbf{j}, a_3\mathbf{k}$

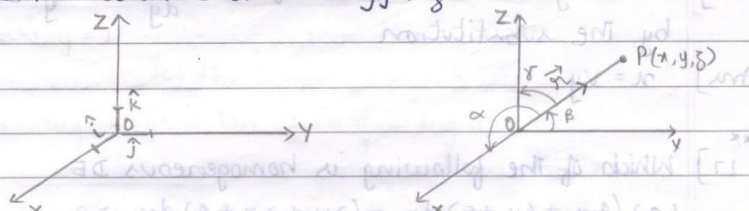
If $\vec{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\vec{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$

then they are position vectors of points A and B, then

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (b_1 - a_1)\mathbf{i} + (b_2 - a_2)\mathbf{j} + (b_3 - a_3)\mathbf{k}$$

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = (a_1 - b_1)\mathbf{i} + (a_2 - b_2)\mathbf{j} + (a_3 - b_3)\mathbf{k}$$

Consider a point $P(x, y, z)$ in a 3D space. Then its position vector is $\vec{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$



- $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = \overrightarrow{OP}$
- $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$
- $\hat{i}$ is unit vector on x-axis, $\hat{j}$ is unit vector on y-axis and $\hat{k}$ is unit vector on z-axis.
- The direction angles of any vector made with the coordinate axes are α, β, γ respectively

cosines of these angles give the direction cosines (DCs) of the given vector and are represented by l, m, n .

Hence, $l = \cos \alpha$, $m = \cos \beta$, $n = \cos \gamma$

- $l^2 + m^2 + n^2 = 1$
 $\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$
- $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$

For a given vector: $\vec{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

$\rightarrow |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

$\rightarrow$ Direction ratios (DRs):
 $a = a_1, b = a_2, c = a_3$

$\rightarrow$ Unit vector $= \hat{a} = \frac{\vec{a}}{|\vec{a}|}$

$\rightarrow$ DCs: $\frac{a_1}{|\vec{a}|}, \frac{a_2}{|\vec{a}|}, \frac{a_3}{|\vec{a}|}$

Eg: If $\vec{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\vec{b} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\vec{c} = 2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$

(i) $|\vec{a}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$
 $|\vec{b}| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$ $|(1, 2, 2)| = 3$
 $|\vec{c}| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$ $|(2, 3, 6)| = 7$

(ii) $3\vec{a} - 2\vec{c} = 3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} - (4\mathbf{i} + 6\mathbf{j} - 12\mathbf{k})$
 $= -\mathbf{i} - 3\mathbf{j} + 15\mathbf{k}$

(iii) Unit vector of $\vec{c} - \vec{a}$
 $\vec{c} - \vec{a} = \mathbf{i} + 2\mathbf{j} - 7\mathbf{k}$
 Unit vect of $\vec{c} - \vec{a} = \frac{\vec{c} - \vec{a}}{|\vec{c} - \vec{a}|} = \frac{\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}}{\sqrt{1 + 4 + 49}} = \frac{\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}}{\sqrt{54}}$

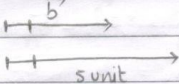
DC's are $\equiv \frac{1}{\sqrt{54}}, \frac{2}{\sqrt{54}}, \frac{-7}{\sqrt{54}}$

(iv) Find unit vector of $\vec{a}$ and hence its DC's
 Unit vector of $\vec{a} = \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\mathbf{i} + \mathbf{j} + \mathbf{k}}{\sqrt{3}}$

DC's are: $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Since the DC's are all equal, the vector $\vec{i} + \vec{j} + \vec{k}$ is equally inclined to all the coordinate axes i.e., $\alpha = \beta = \gamma$

(v) Find a vector in the direction of $\vec{b}$ of magnitude 5.



Vector along $\vec{b}$ of magnitude 5 = $5 \hat{b}$

$$= 5 \frac{\vec{b}}{|\vec{b}|} = 5 \frac{(2\vec{i} - \vec{j} + 2\vec{k})}{\sqrt{4+1+4}}$$

$$= \frac{10\vec{i} - 5\vec{j} + 10\vec{k}}{3}$$

** Types of vectors:

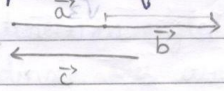
Zero vector: A vector whose initial and terminal points coincide is called a zero vector or a null vector and is denoted as $\vec{0}$. It may be regarded as having any direction.

$\vec{AA}, \vec{BB}$ represent $\vec{0}$.

Unit vector: A vector whose magnitude is unity (1 unit) is called a unit vector. The unit vector in the direction of $\vec{a}$ is denoted by $\hat{a}$.

Coinitial vectors: Two or more vectors having the same initial points are called coinitial vectors.

** Collinear vectors: Two or more vectors are said to be collinear if they are parallel to the same line, irrespective of their magnitudes and directions.



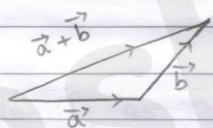
Parallel vectors:

****** Negative of a vector: A vector whose magnitude is same as that of a given vector but has opposite direction of given vector, is called negative of the given vector.

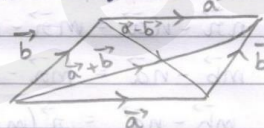
If given vector is $\vec{AB}$ then its negative vector is $\vec{BA}$ and is written as $\vec{BA} = -\vec{AB}$.

Equal vectors: Two vectors $\vec{A}$ and $\vec{B}$ are said to be equal if they have same magnitude and direction regardless of positions of their initial points and is written as $\vec{A} = \vec{B}$.

Triangle law of addition:



Parallelogram law of addition:



****** Section formula:

(a) For internal division: $\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$

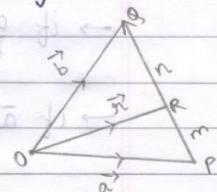
(b) For external division: $\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}$

Let P, Q be two points with PV's $\vec{a}, \vec{b}$ and let point R with PV $\vec{r}$ divide $\vec{PQ}$ in the ratio m:n.

case (i): When point R divides $\vec{PQ}$ internally in ratio m:n.

$$\frac{PR}{RQ} = \frac{m}{n}$$

$$\frac{\vec{OR} - \vec{OP}}{\vec{OQ} - \vec{OR}} = \frac{m}{n}$$



$$\frac{\vec{r} - \vec{a}}{\vec{b} - \vec{a}} = \frac{m}{n}$$

$$n\vec{r} - n\vec{a} = m\vec{b} - m\vec{a}$$

$$m\vec{a} + n\vec{r} = m\vec{b} + n\vec{a}$$

$$\vec{r}(m+n) = m\vec{b} + n\vec{a}$$

$$\boxed{\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}}$$

case (iii): When R divides $\vec{PQ}$ externally in the ratio $m:n$

$$\frac{PR}{QR} = \frac{m}{n}$$

$$\frac{\vec{OR} - \vec{OP}}{\vec{OR} - \vec{OQ}} = \frac{m}{n}$$

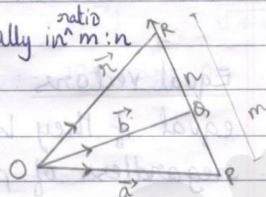
$$\frac{\vec{r} - \vec{a}}{\vec{r} - \vec{b}} = \frac{m}{n}$$

$$n\vec{r} - n\vec{a} = m\vec{r} - m\vec{b}$$

$$m\vec{b} - n\vec{a} = m\vec{r} - n\vec{r}$$

$$m\vec{b} - n\vec{a} = \vec{r}(m-n)$$

$$\boxed{\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}}$$



NOTE: * $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ [vector addition is commutative]

* $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ [vector addition is associative]

* zero vector is identity element under vector addition

i.e. $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$

** * Consider two vectors:

$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$; $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ then

→ if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \Rightarrow \vec{a} \parallel \vec{b}$

→ if $\vec{a} = \lambda\vec{b} \Rightarrow \vec{a} \parallel \vec{b}$ and $\vec{a}, \vec{b}$ are collinear vectors.

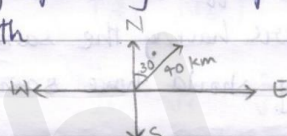
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✳ In the vector : $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

- a_1, a_2, a_3 are direction ratios of $\vec{a}$ and are called scalar components.
- $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ = magnitude of $\vec{a}$
- $\frac{a_1}{|\vec{a}|}, \frac{a_2}{|\vec{a}|}, \frac{a_3}{|\vec{a}|}$ are DC's of $\vec{a}$
- Unit vector : $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$

EXERCISE 10.1

1] Represent graphically a displacement of 40 km, 30° east of north



2] Classify as scalars and vectors:

- 10 kg - weight - scalar
- 2 m, north-west - displacement - vector
- 40° - temperature - scalar
- 40 W - power - scalar
- 10^{-19} C - charge - scalar
- 20 m/s^2 - acceleration - vector

3] Classify as scalars and vectors:

- Time period - scalar
- Distance - scalar
- Force - vector
- Velocity - vector
- Work done - scalar

$\vec{A} = \frac{1}{|\vec{A}|} \vec{A} = \frac{1}{\sqrt{A_x^2 + A_y^2 + A_z^2}} (A_x\hat{i} + A_y\hat{j} + A_z\hat{k})$

4] Identify the vectors in diagram:

i] coinitial vectors: $\vec{a}$ & $\vec{d}$

ii] equal vectors: $\vec{b}$ & $\vec{c}$

iii] collinear but not equal: $\vec{a}, \vec{c} \because \vec{a} = -\vec{c}$

5] True or False:

i] Vector $\vec{a}$ and $-\vec{a}$ are collinear - T

ii] Two collinear vectors are always equal in magnitude - F

Eg: $\vec{a} = i + j + k$; $\vec{b} = 2i + 2j + 2k \Rightarrow \vec{b} = 2\vec{a}$
 $\therefore \vec{a} \parallel \vec{b} \Rightarrow \vec{a}, \vec{b}$ are collinear with different magnitudes

iii] Two vectors having same magnitude are collinear - F
 $\therefore$ They may have different directions.

iv] Two collinear vectors having the same magnitude are equal - F ($\because$ they should have same direction also)

EXERCISE 10.2

Eg 4] Find the values of x, y, z so that the vectors
 $\vec{a} = xi + 2j + zk$ and $\vec{b} = 2i + yj + k$ are equal.

Sol] $x = 2, y = 2, z = 1$

Eg 5] Let $\vec{a} = i + 2j$, $\vec{b} = 2i + j - 3k$ are $|\vec{a}| = |\vec{b}|$, are
 $\vec{a}$ and $\vec{b}$ equal?

Sol] $|\vec{a}| = \sqrt{1+4} = \sqrt{5}$ } $\Rightarrow |\vec{a}| = |\vec{b}| = \sqrt{5}$
 $|\vec{b}| = \sqrt{4+1+9} = \sqrt{14}$
 $\vec{a} \neq \vec{b}$: their components are different.

Eg 6] Find unit vector in the direction of
 $\vec{a} = 2i + 3j + k$

unit vector: $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{2i + 3j + k}{\sqrt{4+9+1}} = \frac{2i + 3j + k}{\sqrt{14}}$
 $= \frac{2}{\sqrt{14}}i + \frac{3}{\sqrt{14}}j + \frac{1}{\sqrt{14}}k$

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**** Eg 7]** Find a vector in the direction of $\vec{a} = i - 2j$ that has magnitude 7 units.

Sol] Vector along $\vec{a}$ of magnitude 7 $= 7\hat{a} = 7 \frac{\vec{a}}{|\vec{a}|}$

$$= \frac{7(i-2j)}{\sqrt{1+4}} = \frac{7i-14j}{\sqrt{5}} = \left[\frac{7}{\sqrt{5}}i - \frac{14}{\sqrt{5}}j \right]$$

Eg 9] Write the D.R's of $\vec{a} = i + j - 2k$ and hence calculate its D.C's.

Sol] $|\vec{a}| = \sqrt{1+1+4} = \sqrt{6}$

(i) D.R's are : $a=1, b=1, c=-2$

(ii) D.C's are : $l = \frac{a}{|\vec{a}|} = \frac{1}{\sqrt{6}}, m = \frac{b}{|\vec{a}|} = \frac{1}{\sqrt{6}}, n = \frac{c}{|\vec{a}|} = \frac{-2}{\sqrt{6}}$

$\Rightarrow$ D.C's are $\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$

Eg 8] Find unit vector in the direction of sum of the vectors $\vec{a} = 2i + 2j - 5k, \vec{b} = 2i + j + 3k$.

Sol] Sum $= \vec{a} + \vec{b}$

$$= 4i + 3j - 2k$$

Unit vector of $\vec{a} + \vec{b} = \frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} = \frac{4i + 3j - 2k}{\sqrt{16+9+4}} = \frac{4i + 3j - 2k}{\sqrt{29}}$

(Im) Eg 10] Find vector joining the points $P(2, 3, 0)$ and $Q(-1, -2, -4)$ directed from P to Q.

Sol] $\vec{PQ} = \vec{OQ} - \vec{OP} = -3i - 5j - 4k$

Eg 11] Consider points P, Q with P.V's $\vec{OP} = 3\vec{a} - 2\vec{b}$ and $\vec{OQ} = \vec{a} + \vec{b}$. Find P.V of R that divides $\vec{PQ}$ in the ratio 2:1 (i) internally (ii) externally.

Sol] (i) For internal division

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

$$\vec{r} = \frac{2(\vec{a} + \vec{b}) + 1(3\vec{a} - 2\vec{b})}{2+1} = \frac{5\vec{a} + 0\vec{b}}{3} = \frac{5\vec{a}}{3}$$

(ii) For external division: $\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}$

$$\vec{r} = \frac{2(\vec{a} + \vec{b}) - 1(3\vec{a} - 2\vec{b})}{2-1} = \frac{-\vec{a} + 4\vec{b}}{1} = -\vec{a} + 4\vec{b}$$

(2m) Eg 12] ST the points $A(2i-j+k)$, $B(i-3j-5k)$, $C(3i-4j-9k)$ are vertices of a right angle Δ^e .

Sol] $\vec{AB} = \vec{OB} - \vec{OA} = -i - 2j - 6k$; $|\vec{AB}| = \sqrt{1+4+36} = \sqrt{41}$

$$\vec{BC} = \vec{OC} - \vec{OB} = 2i - j + k$$
; $|\vec{BC}| = \sqrt{4+1+1} = \sqrt{6}$

$$\vec{CA} = \vec{OA} - \vec{OC} = -i + 3j + 5k$$
; $|\vec{CA}| = \sqrt{1+9+25} = \sqrt{35}$

$$\text{Clearly, } (\sqrt{35})^2 + (\sqrt{6})^2 = (\sqrt{41})^2$$

$$\Rightarrow |\vec{CA}|^2 + |\vec{BC}|^2 = |\vec{AB}|^2$$

$$\Rightarrow ABC \text{ is a right angle } \Delta^e$$

17] ST the points A, B, C with PV's $\vec{OA} = 3i - 4j - 9k$,

$\vec{OB} = 2i - j + k$, $\vec{OC} = i - 3j - 5k$ form the vertices of Δ^e .

$$\vec{AB} = \vec{OB} - \vec{OA} = -i + 3j + 5k$$
; $|\vec{AB}| = \sqrt{1+9+25} = \sqrt{35}$

$$\vec{BC} = \vec{OC} - \vec{OB} = -i - 2j - 6k$$
; $|\vec{BC}| = \sqrt{1+4+36} = \sqrt{41}$

$$\vec{CA} = \vec{OA} - \vec{OC} = 2i - j + k$$
; $|\vec{CA}| = \sqrt{4+1+1} = \sqrt{6}$

$$\text{Clearly, } (\sqrt{35})^2 + (\sqrt{6})^2 = (\sqrt{41})^2$$

$$\Rightarrow |\vec{AB}|^2 + |\vec{CA}|^2 = |\vec{BC}|^2$$

$$\Rightarrow ABC \text{ is a right angle } \Delta^e$$

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- 1] Compute the mag of vectors:
 - i] $\vec{a} = i + j + k \Rightarrow |\vec{a}| = \sqrt{1+1+1} = \sqrt{3}$
 - ii] $\vec{b} = 2i - 7j - 3k \Rightarrow |\vec{b}| = \sqrt{4+49+9} = \sqrt{62}$
 - iii] $\vec{c} = \frac{1}{\sqrt{3}}i + \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k \Rightarrow |\vec{c}| = \sqrt{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}} = 1$
- 2] Write two different vectors having same magnitude.
 $\vec{a} = 2i - j$ and $\vec{b} = i + 2j$
- 3] Write two different vectors with same direction.
 $\vec{a} = i + 2j + 3k$ and $\vec{b} = 2i + 4j + 6k$
 $\vec{a} = \lambda \vec{b}$
- 4] Find the values of x, y so that the vectors $\vec{a} = 2i + 3j$ and $\vec{b} = xi + yj$ are equal.
 $x = 2, y = 3$
- 5] Find the scalar and vector components of vector with initial point $(2, 1)$ and terminal point $(-5, 7)$.
 sol] $\vec{AB} = \vec{OB} - \vec{OA} = -7i + 6j$
 Scalar components are: $-7, 6$
 Vector " " : $-7i, 6j$
- 6] Find the sum of the vectors $\vec{a} = i - 2j + k, \vec{b} = -2i + 4j + 5k,$
 $\vec{c} = i - 6j - 7k$
 Sum = $\vec{a} + \vec{b} + \vec{c} = 0i - 4j - k = -4j - k$
- 7] Find the unit vector in the direction of $\vec{a} = i + j + 2k$.
 $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{i + j + 2k}{\sqrt{1+1+4}} = \frac{1}{\sqrt{6}}i + \frac{1}{\sqrt{6}}j + \frac{2}{\sqrt{6}}k$

8] Find the unit vector in the direction of $\vec{PQ}$ with $P(1, 2, 3)$, $Q(4, 5, 6)$

Sol] $\vec{PQ} = \vec{OQ} - \vec{OP} = 3\vec{i} + 3\vec{j} + 3\vec{k} = 3(\vec{i} + \vec{j} + \vec{k})$
 unit vector along $\vec{PQ} = \frac{\vec{PQ}}{|\vec{PQ}|} = \frac{3(\vec{i} + \vec{j} + \vec{k})}{3\sqrt{1+1+1}} = \frac{1}{\sqrt{3}}\vec{i} + \frac{1}{\sqrt{3}}\vec{j} + \frac{1}{\sqrt{3}}\vec{k}$

9] ~~Let~~ $\vec{a} = 2\vec{i} - \vec{j} + 2\vec{k}$, $\vec{b} = -\vec{i} + \vec{j} - \vec{k}$ find the unit vector in the direction of $\vec{a} + \vec{b}$.

Sol] $\vec{a} + \vec{b} = \vec{i} + \vec{k}$
 unit vector of $\vec{a} + \vec{b} = \frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} = \frac{\vec{i} + \vec{k}}{\sqrt{2}} = \frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{k}$

10] Find a vector in the direction of $5\vec{i} - \vec{j} + 2\vec{k}$ which has magnitude 8 units.

Sol] Let $\vec{a} = 5\vec{i} - \vec{j} + 2\vec{k}$
 vector along $\vec{a}$ along $= 8\hat{a} = \frac{8|\vec{a}|}{|\vec{a}|} = \frac{8(5\vec{i} - \vec{j} + 2\vec{k})}{\sqrt{25+1+4}}$
 $= \frac{40}{\sqrt{30}}\vec{i} - \frac{8}{\sqrt{30}}\vec{j} + \frac{16}{\sqrt{30}}\vec{k}$

11] ST the vectors $\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$ and $\vec{b} = -4\vec{i} + 6\vec{j} - 8\vec{k}$ are collinear.

Clearly, $\vec{b} = -2(\vec{a}) \Rightarrow \vec{a}, \vec{b}$ are collinear vectors.

(or) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \frac{-1}{2} \Rightarrow \vec{a} \parallel \vec{b}$ or $\vec{a}, \vec{b}$ are collinear?

12] Find the DC's of the vector $\vec{i} + 2\vec{j} + 3\vec{k}$.
 $|\vec{a}| = \sqrt{1+4+9} = \sqrt{14}$

DC's are: $l = \frac{1}{\sqrt{14}}$, $m = \frac{2}{\sqrt{14}}$, $n = \frac{3}{\sqrt{14}}$

- 13] Find DC's of the vector joining $A(1, 2, -3)$ and $B(-1, -2, 1)$ directed from A to B.

Sol] $\vec{AB} = \vec{OB} - \vec{OA} = -2\hat{i} - 4\hat{j} + 4\hat{k} = \vec{a} = \vec{b}$ [b2]

$$\vec{AB} = 2(-\hat{i} - 2\hat{j} + 2\hat{k})$$

$$|\vec{AB}| = 2\sqrt{1+4+4} = 6$$

$$\text{DC's are } \frac{-2}{6}, \frac{-4}{6}, \frac{4}{6} \text{ i.e., } \frac{-1}{3}, \frac{-2}{3}, \frac{2}{3}$$

- 14] ST the vector $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the axis OX , OY and OZ . Gn:

Sol] Gn: $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$$|\vec{a}| = \sqrt{1+1+1} = \sqrt{3}$$

$$\text{DC's are : } l = \frac{1}{\sqrt{3}}, m = \frac{1}{\sqrt{3}}, n = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \text{DC's are all equal i.e. } \alpha = \beta = \gamma$$

$\therefore$ The gn. vector is equally inclined to the coordinate axes.

- 15] Find the PV of pt R which divides line joining P and Q with PV's $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respec. in the ratio 2:1 (a) internally (b) externally.

Sol] Gn: $\vec{OP} = \vec{a} = \hat{i} + 2\hat{j} - \hat{k}$; $\vec{OQ} = \vec{b} = -\hat{i} + \hat{j} + \hat{k}$; $m:n = 2:1$

(i) Internal division:

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n} = \frac{2(-\hat{i} + \hat{j} + \hat{k}) + 1(\hat{i} + 2\hat{j} - \hat{k})}{2+1}$$

$$\vec{r} = \frac{-\hat{i} + 4\hat{j} + \hat{k}}{3}$$

(ii) Externally:

$$\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n} = \frac{2(-\hat{i} + \hat{j} + \hat{k}) - 1(\hat{i} + 2\hat{j} - \hat{k})}{2-1}$$

$$\vec{r} = \frac{-3\hat{i} + 3\hat{k}}{1}$$

16] Find the PV of mid point of the vector joining $P(2, 3, 4)$ and $Q(4, 1, -2)$.

Sol] Giv: $\vec{a} = \vec{OP} = 2\hat{i} + 3\hat{j} + 4\hat{k}$
 $\vec{b} = \vec{OQ} = 4\hat{i} + \hat{j} - 2\hat{k}$
 mid point = $\vec{r} = \frac{\vec{a} + \vec{b}}{2} = \frac{6\hat{i} + 4\hat{j} + 2\hat{k}}{2} = 3\hat{i} + 2\hat{j} + \hat{k}$

Product of Two vectors:

SCALAR PRODUCT OR DOT PRODUCT:

* Dot product of two ~~para~~ vectors gives a constant and hence called scalar product.

* $\forall \vec{a}, \vec{b} \in V, \vec{a} \cdot \vec{b} \notin V$

$\therefore$ dot product is not binary operation on set of vectors V
 i.e., ' V ' is not closed under ' $\cdot$ '.

* $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (dot product is commutative)

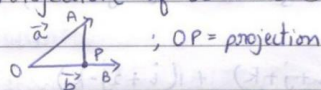
* $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

* $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$

* $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

* $\vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$

* Projection of $\vec{a}$ onto $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \vec{a} \cdot \hat{b}$



* $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$

* $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$

* $\vec{a} \cdot \vec{a} = |\vec{a}|^2$

Eg 13] Find the angle b/w 2 vectors $\vec{a}$ and $\vec{b}$ with magnitudes 1 and 2 when $\vec{a} \cdot \vec{b} = 1$

$$\text{Sol]} \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{1}{1 \cdot 2} = \frac{1}{2} = \cos 60^\circ$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

Eg 14] Find θ b/w the vectors $\vec{a} = i + j - k$, $\vec{b} = i - j + k$

$$\text{Sol]} \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{(i + j - k) \cdot (i - j + k)}{\sqrt{1+1+1} \cdot \sqrt{1+1+1}}$$

$$\cos \theta = \frac{1 - 1 - 1}{\sqrt{3} \cdot \sqrt{3}} = -\frac{1}{3}$$

$$\Rightarrow \theta = \cos^{-1}\left(-\frac{1}{3}\right)$$

Eg 15] If $\vec{a} = 5i - j - 3k$, $\vec{b} = i + 3j - 5k$. ST the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are $\perp$.

$$\text{Sol]} \text{Gn: } \vec{a} = 5i - j - 3k; \vec{b} = i + 3j - 5k$$

$$\vec{a} + \vec{b} = 6i + 2j - 8k$$

$$\vec{a} - \vec{b} = 4i - 4j + 2k$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 24 - 8 - 16 = 24 - 24 = 0$$

$$\Rightarrow (\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$$

Eg 16] Find the projection of $\vec{a} = 2i + 3j + 2k$ on the $\vec{b} = i + 2j + k$

$$\text{Sol]} \text{Projection of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$= \frac{(2i + 3j + 2k) \cdot (i + 2j + k)}{\sqrt{1+4+1}}$$

$$= \frac{2 + 6 + 2}{\sqrt{6}} = \frac{10}{\sqrt{6}} \text{ units}$$

Eg 17] Find $|\vec{a} - \vec{b}|$ if two vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 4$

Sol] Consider $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$
 $= 2^2 + 3^2 - 2(4)$
 $= 4 + 9 - 8 = 5$
 $\therefore |\vec{a} - \vec{b}| = \sqrt{5}$

Eg 18] If $\vec{a}$ is a unit vector and $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 8$. Find $|\vec{x}|$

Sol] Gn: $|\vec{a}| = 1$
 $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 8$
 $|\vec{x}|^2 - |\vec{a}|^2 = 8$
 $|\vec{x}|^2 - 1 = 8$
 $|\vec{x}|^2 = 9$
 $|\vec{x}| = 3$

Eg 21] ST the points $A(-2\hat{i} + 3\hat{j} + 5\hat{k})$, $B(\hat{i} + 2\hat{j} + 3\hat{k})$, $C(7\hat{i} - \hat{k})$ are collinear.

Sol] $\vec{AB} = \vec{OB} - \vec{OA} = 3\hat{i} - \hat{j} - 2\hat{k}$; $|\vec{AB}| = \sqrt{9+1+4} = \sqrt{14}$
 $\vec{BC} = \vec{OC} - \vec{OB} = 6\hat{i} - 2\hat{j} - 4\hat{k}$; $|\vec{BC}| = \sqrt{36+4+16} = 2\sqrt{14}$
 $\vec{CA} = \vec{OA} - \vec{OC} = -9\hat{i} + 3\hat{j} + 6\hat{k}$; $|\vec{CA}| = \sqrt{81+9+36} = 3\sqrt{14}$
 Clearly, $\sqrt{14} + 2\sqrt{14} = 3\sqrt{14}$
 $\Rightarrow |\vec{AB}| + |\vec{BC}| = |\vec{CA}|$
 $\Rightarrow$ gn. points are collinear.

(OR)
 DR's of $\vec{AB}$ and $\vec{BC}$ are proportional
 $\Rightarrow \vec{AB} \parallel \vec{BC}$ but pt. 'B' is common
 $\therefore$ The gn. pts are collinear.

*] If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$ p.T. $\vec{a}$ & $\vec{b}$ are $\perp$

Sol] sbs $|\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2$
 $|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}$
 $4\vec{a} \cdot \vec{b} = 0$
 $\vec{a} \cdot \vec{b} = 0$
 $\Rightarrow \vec{a} \perp \vec{b}$

EXERCISE 10.3

- 1) Find the angle b/w 2 vectors $\vec{a}$ and $\vec{b}$ with mag. $\sqrt{3}$ & 2, and $\vec{a} \cdot \vec{b} = \sqrt{6}$

Sol] Gn: $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 2$, $\vec{a} \cdot \vec{b} = \sqrt{6}$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{\sqrt{6}}{\sqrt{3} \cdot 2} = \frac{\sqrt{3} \sqrt{2}}{\sqrt{3} \cdot 2} = \frac{1}{\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

- 2) Find the angle b/w the vectors $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$ and $\vec{b} = 3\vec{i} - 2\vec{j} + \vec{k}$

Sol] Gn: $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$, $\vec{b} = 3\vec{i} - 2\vec{j} + \vec{k}$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{3 + 4 + 3}{\sqrt{1+4+9} \sqrt{9+4+1}} = \frac{10}{14} = \frac{5}{7}$$

$$\theta = \cos^{-1}\left(\frac{5}{7}\right)$$

- 3) Find the projection of the vector $\vec{a} = \vec{i} - \vec{j}$ on $\vec{b} = \vec{i} + \vec{j}$

Sol] Gn: $\vec{a} = \vec{i} - \vec{j}$, $\vec{b} = \vec{i} + \vec{j}$

$$\text{Proj of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{1-1}{\sqrt{2}} = 0$$

- 4) Find the proj. of the vector $\vec{a} = \vec{i} + 3\vec{j} + 7\vec{k}$ on $\vec{b} = 7\vec{i} - \vec{j} + 8\vec{k}$

Sol] Gn: $\vec{a} = \vec{i} + 3\vec{j} + 7\vec{k}$, $\vec{b} = 7\vec{i} - \vec{j} + 8\vec{k}$

$$\text{Proj of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{7 - 3 + 56}{\sqrt{49 + 1 + 64}} = \frac{60}{\sqrt{114}}$$

- 5) If each of the 3 vectors is a unit vector, also show that they are mutually $\perp$ to each other.

$$\frac{1}{7}(2\vec{i} + 3\vec{j} + 6\vec{k}), \quad \frac{1}{7}(3\vec{i} - 6\vec{j} + 2\vec{k}), \quad \frac{1}{7}(6\vec{i} + 2\vec{j} - 3\vec{k})$$

$$\vec{a}, \quad \vec{b}, \quad \vec{c}$$

$$|\vec{a}| = \frac{1}{7} \sqrt{4+9+36} = \frac{\sqrt{49}}{7} = \frac{7}{7} = 1 \Rightarrow \vec{a} \text{ is unit vector}$$

$$|\vec{b}| = \frac{1}{7} \sqrt{9+36+9} = \frac{\sqrt{49}}{7} = \frac{7}{7} = 1 \Rightarrow \vec{b} \text{ is unit vector}$$

$$|\vec{c}| = \frac{1}{7} \sqrt{36+4+9} = \frac{\sqrt{49}}{7} = \frac{7}{7} = 1 \Rightarrow \vec{c} \text{ is unit vector}$$

$$\text{Consider } \vec{a} \cdot \vec{b} = \frac{1}{49} (6-18+12) = \frac{18-18}{49} = 0 \Rightarrow \vec{a} \perp \vec{b} \rightarrow \textcircled{1}$$

$$\vec{b} \cdot \vec{c} = \frac{1}{49} (18-12-6) = 0 \Rightarrow \vec{b} \perp \vec{c} \rightarrow \textcircled{2}$$

$$\vec{c} \cdot \vec{a} = \frac{1}{49} (12+18-18) = 0 \Rightarrow \vec{c} \perp \vec{a} \rightarrow \textcircled{3}$$

$\textcircled{1}, \textcircled{2}, \textcircled{3} \Rightarrow \vec{a}, \vec{b}, \vec{c} \text{ are mutually } \perp^{\text{le}}$

6) $|\vec{a}|, |\vec{b}|$ if $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$ and $|\vec{a}| = 8|\vec{b}|$

Sol] $\text{Gn: } (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$

$$|\vec{a}|^2 - |\vec{b}|^2 = 8$$

$$(8|\vec{b}|)^2 - |\vec{b}|^2 = 8$$

$$64|\vec{b}|^2 - |\vec{b}|^2 = 8$$

$$63|\vec{b}|^2 = 8$$

$$|\vec{b}| = \sqrt{\frac{8}{63}} \text{ units}$$

$$\begin{aligned} |\vec{a}| &= 8|\vec{b}| \\ &= 8 \sqrt{\frac{8}{63}} \text{ units} \end{aligned}$$

7] Evaluate $(3\vec{a} - 5\vec{b}) \cdot (2\vec{a} + 7\vec{b})$

Sol] $= 6(\vec{a} \cdot \vec{a}) + 21(\vec{a} \cdot \vec{b}) - 10(\vec{b} \cdot \vec{a}) - 35(\vec{b} \cdot \vec{b})$

$$= 6|\vec{a}|^2 + 21(\vec{a} \cdot \vec{b}) - 10(\vec{a} \cdot \vec{b}) - 35|\vec{b}|^2$$

$$= 6|\vec{a}|^2 + 11(\vec{a} \cdot \vec{b}) - 35|\vec{b}|^2$$

8] Find mag. of 2 vectors $\vec{a}$ and $\vec{b}$ having the same mag such that angle b/w them is 60° and their scalar product is half.

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Sol] $|\vec{a}| = |\vec{b}|$, $\theta = 60^\circ$, $\vec{a} \cdot \vec{b} = \frac{1}{2} + \sqrt{3} + \sqrt{3}$
 $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{1 + 2\sqrt{3}}{2}$
 $\cos 60^\circ = \frac{1}{2} = \frac{1 + 2\sqrt{3}}{2}$
 $\frac{1}{2} = \frac{1 + 2\sqrt{3}}{2}$
 $1 = 1 + 2\sqrt{3}$
 $0 = 2\sqrt{3}$
 $\sqrt{3} = 0$
 $\sqrt{3} = 0$

9) Find $|\vec{x}|$ if for a unit vector $\vec{a}$ $(\vec{x} - \vec{a})(\vec{x} + \vec{a}) = 12$
 Sol] Gn: $|\vec{a}| = 1$
 $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 12$
 $|\vec{x}|^2 - |\vec{a}|^2 = 12$
 $|\vec{x}|^2 - 1 = 12$
 $|\vec{x}|^2 = 13$
 $|\vec{x}| = \sqrt{13}$

** 10) If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{c} = 3\hat{i} + \hat{j}$ such that $(\vec{a} + \lambda \vec{b}) \perp \vec{c}$. Find λ .
 Sol] $(\vec{a} + \lambda \vec{b}) \perp \vec{c} \Rightarrow (\vec{a} + \lambda \vec{b}) \cdot \vec{c} = 0$
 $[(2\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(-\hat{i} + 2\hat{j} + \hat{k})] \cdot (3\hat{i} + \hat{j}) = 0$
 $[(2 - \lambda)\hat{i} + (2 + 2\lambda)\hat{j} + (3 + \lambda)\hat{k}] \cdot (3\hat{i} + \hat{j}) = 0$
 $3(2 - \lambda) + 1(2 + 2\lambda) = 0$
 $6 - 3\lambda + 2 + 2\lambda = 0$
 $8 - \lambda = 0$
 $\lambda = 8$

** 13] If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$. Find the value of $(\vec{a} \cdot \vec{b}) + (\vec{b} \cdot \vec{c}) + (\vec{c} \cdot \vec{a})$.
 Sol] Gn: $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$
 $\vec{a} + \vec{b} + \vec{c} = 0$
 $(\vec{a} + \vec{b} + \vec{c})^2 = 0$ (sbs)

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0 \quad (\text{b2})$$

$$(1+1+1) + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0 \quad \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

- 16] ST the points $A(1, 2, 7)$, $B(2, 6, 3)$, $C(3, 10, -1)$ are collinear
 $\vec{AB} = \vec{OB} - \vec{OA} = \hat{i} + 4\hat{j} - 4\hat{k}$; $|\vec{AB}| = \sqrt{1+16+16} = \sqrt{33}$
 $\vec{BC} = \vec{OC} - \vec{OB} = \hat{i} + 4\hat{j} - 4\hat{k}$; $|\vec{BC}| = \sqrt{1+16+16} = \sqrt{33}$
 $\vec{CA} = \vec{OA} - \vec{OC} = 2\hat{i} + 8\hat{j} - 8\hat{k}$; $|\vec{CA}| = \sqrt{4+64+64} = 2\sqrt{33}$
 $\therefore \sqrt{33} + \sqrt{33} = 2\sqrt{33}$
 $|\vec{AB}| + |\vec{BC}| = |\vec{CA}|$
 $\Rightarrow$ Gn pts are collinear.

- 17] ST the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ form the vertices of a Δ .
 $|\vec{AB}| = \vec{OB} - \vec{OA} = -\hat{i} - 2\hat{j} - 6\hat{k}$; $|\vec{AB}| = \sqrt{1+4+36} = \sqrt{41}$
 $|\vec{BC}| = \vec{OC} - \vec{OB} = 2\hat{i} - \hat{j} + \hat{k}$; $|\vec{BC}| = \sqrt{4+1+1} = \sqrt{6}$
 $|\vec{CA}| = \vec{OA} - \vec{OC} = -\hat{i} + 3\hat{j} + 5\hat{k}$; $|\vec{CA}| = \sqrt{1+9+25} = \sqrt{35}$
 $(\sqrt{35})^2 + (\sqrt{6})^2 = (\sqrt{41})^2$
 $\Rightarrow |\vec{BC}|^2 + |\vec{CA}|^2 = |\vec{AB}|^2$
 $\Rightarrow A, B, C$ are vertices of Δ .

- (im) ** 18] If $\vec{a}$ is a non zero vector of magnitude $|\vec{a}|$ and λ a non-zero scalar, then $\lambda\vec{a}$ is a unit vector if

(a) $\lambda = 1$ (b) $\lambda = -1$, (c) $\lambda = \frac{1}{|\vec{a}|}$ ~~(d) $\lambda = \frac{1}{|\vec{a}|}$~~

$|\vec{a}| = a$, $\lambda\vec{a}$ is unit vector

$0 = 5 + 5 + 5 = 15 \Rightarrow |\lambda\vec{a}| = 1$

$(\vec{a} \cdot \vec{a}) (\lambda \lambda) = 1 \cdot 1 = 1$

$|\lambda| a = 1 \Rightarrow |\lambda| = \frac{1}{a} = \frac{1}{|\vec{a}|}$

$a = \frac{1}{|\lambda|}$

$0 = 5 + 5 + 5 = 15$

CROSS PRODUCT / VECTOR PRODUCT:

* Cross product of two vectors gives a vector and hence called vector product.

* $\forall \vec{a}, \vec{b} \in V, (\vec{a} \times \vec{b}) \in V$.

$\therefore$ 'X' product is binary operation on V

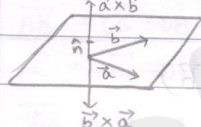
Also 'V' is closed under 'X' product.

* $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$

* $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \hat{n}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

* $\vec{a} \times \vec{b}$ is a vector $\perp$ to both $\vec{a}$ and $\vec{b}$



• Here vector $\vec{a} \times \vec{b}$ is $\perp$ to entire plane and hence $\perp$ to all the vectors present in the plane.

• $\hat{n}$ is unit vector on $\vec{a} \times \vec{b}$

* $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$

* $\vec{a} \times \vec{a} = 0$

* $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$

* $\hat{i} \times \hat{j} = \hat{k}; \hat{j} \times \hat{k} = \hat{i}; \hat{k} \times \hat{i} = \hat{j}$

$\hat{j} \times \hat{i} = -\hat{k}; \hat{k} \times \hat{j} = -\hat{i}; \hat{i} \times \hat{k} = -\hat{j}$

* $|\vec{a} \times \vec{b}| = \text{An. of parallelogram}$

(where $\vec{a}$ and $\vec{b}$ are adjacent sides)

* $\frac{1}{2}|\vec{a} \times \vec{b}| = \text{An. of triangle}$

* $\frac{1}{2}|\vec{AB} \times \vec{AC}| = \text{An. of triangle (Gn. vertices A, B, C)}$

EXERCISE 10.4

Eg 22] Find $|\vec{a} \times \vec{b}|$ if $\vec{a} = 2\vec{i} + \vec{j} + 3\vec{k}$, $\vec{b} = 3\vec{i} + 5\vec{j} - 2\vec{k}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 3 \\ 3 & 5 & -2 \end{vmatrix} = \vec{i}(-17) - \vec{j}(-13) + \vec{k}(7) \\ = -17\vec{i} + 13\vec{j} + 7\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{289 + 169 + 49} = \sqrt{507}$$

** Eg 23] Find a unit vector $\perp^e$ to each of the vectors $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$

$$\text{Gn: } \vec{a} = \vec{i} + \vec{j} + \vec{k}, \quad \vec{b} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$\text{Sol] } \vec{a} + \vec{b} = 2\vec{i} + 3\vec{j} + 4\vec{k}$$

$$\vec{a} - \vec{b} = 0\vec{i} - \vec{j} - 2\vec{k}$$

Vector $\perp^e$ to both $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ is $(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 4 \\ 0 & -1 & -2 \end{vmatrix} = \vec{i}(-6+4) - \vec{j}(-4) + \vec{k}(-2) \\ = -2\vec{j} + 4\vec{j} - 2\vec{k} \\ = 2(\vec{i} + 2\vec{j} - \vec{k})$$

$$\text{Its unit vector} = \frac{2(\vec{i} + 2\vec{j} - \vec{k})}{2\sqrt{1+4+1}} = \frac{\vec{i} + 2\vec{j} - \vec{k}}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{6}}\vec{i} + \frac{2}{\sqrt{6}}\vec{j} - \frac{1}{\sqrt{6}}\vec{k}$$

Eg 24] Find area of Δ^e having pts A(1,1,1), B(1,2,3), C(2,3,1) as its vertices.

$$\text{Sol] } \vec{AB} = \vec{OB} - \vec{OA} = 0\vec{i} + \vec{j} + 2\vec{k}$$

$$\vec{AC} = \vec{OC} - \vec{OA} = \vec{i} + 2\vec{j} + 0\vec{k}$$

$$\text{Consider } \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{vmatrix} = \vec{i}(-4) - \vec{j}(-2) + \vec{k}(-1) \\ = -4\vec{i} + 2\vec{j} - \vec{k}$$

$$\text{An of } \Delta^{\text{ic}} = \frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \sqrt{16+9+1} = \frac{\sqrt{26}}{2} \text{ sq. units}$$

Ex 25] Find an of llgm whose adjacent sides are $\vec{a} = 3\vec{i} + \vec{j} + 4\vec{k}$, $\vec{b} = \vec{i} - \vec{j} + \vec{k}$.

$$\text{Sol]} \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 4 \\ 1 & -1 & 1 \end{vmatrix} = \vec{i}(5) - \vec{j}(-1) + \vec{k}(-4) = 5\vec{i} + \vec{j} - 4\vec{k}$$

$$\text{An of llgm} = |\vec{a} \times \vec{b}| = \sqrt{25+1+16} = \sqrt{42} \text{ sq. units}$$

9] Find an of Δ^{ic} with vertices A(1, 1, 2), B(2, 3, 5), C(1, 5, 5).

$$\text{Sol]} \text{An of } \Delta^{\text{ic}} = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} = \vec{OB} - \vec{OA} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$\vec{AC} = \vec{OC} - \vec{OA} = 0\vec{i} + 4\vec{j} + 3\vec{k}$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 3 \\ 0 & 4 & 3 \end{vmatrix} = \vec{i}(-6) - \vec{j}(3) + \vec{k}(4) = -6\vec{i} - 3\vec{j} + 4\vec{k}$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{36+9+16} = \sqrt{60}$$

$$\therefore \text{An of } \Delta^{\text{ic}} = \frac{1}{2} \sqrt{60} = \frac{\sqrt{60}}{2} \text{ sq. units}$$

10] Find an of llgm whose adjacent sides are determined by $\vec{a} = \vec{i} - \vec{j} + 3\vec{k}$, $\vec{b} = 2\vec{i} - 7\vec{j} + \vec{k}$

$$\text{Sol]} \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 3 \\ 2 & -7 & 1 \end{vmatrix} = \vec{i}(20) - \vec{j}(-5) + \vec{k}(-5) = 20\vec{i} + 5\vec{j} - 5\vec{k}$$

$$\text{An of llgm} = |\vec{a} \times \vec{b}| = 5\sqrt{16+1+1} = 5\sqrt{18} \text{ sq. units}$$

12] An of a $\square^{\text{ic}}$ having vertices A, B, C, D with P.V's

$$-\vec{i} + \frac{1}{2}\vec{j} + 4\vec{k}, \vec{i} + \frac{1}{2}\vec{j} + \vec{k}, \vec{i} - \frac{1}{2}\vec{j} + 4\vec{k}, -\vec{i} - \frac{1}{2}\vec{j} + 4\vec{k} \text{ is}$$

$$(a) \frac{1}{2} \quad (b) 1 \quad (c) 2 \quad (d) 4$$

$\vec{AB} = \vec{OB} - \vec{OA} = 2\hat{i} + 0\hat{j} + 0\hat{k} = 2\hat{i}$
 $\vec{AC} = \vec{OC} - \vec{OA} = 0\hat{i} - \hat{j} + 0\hat{k} = -\hat{j}$

An of Δ = An of Δ = $|\vec{AB} \times \vec{AC}|$
 $\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 0 & -1 & 0 \end{vmatrix} = 0\hat{i} + \hat{j}(0) + \hat{k}(-2)$
 $= -2\hat{k}$

An of Δ = $|\vec{AB} \times \vec{AC}| = \sqrt{4} \text{ sq. units} = 2 \text{ sq. units}$

1) Find $|\vec{a} \times \vec{b}|$ if $\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$
 $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -7 & 7 \\ 3 & -2 & 2 \end{vmatrix} = \hat{i}(0) - \hat{j}(19) + \hat{k}(19)$
 $= 19(\hat{j} + \hat{k})$
 $|\vec{a} \times \vec{b}| = \sqrt{(19)^2 + (19)^2} = 19\sqrt{2}$

2) Find unit vector $\perp$ to each of vector $(\vec{a} + \vec{b})$, $(\vec{a} - \vec{b})$.
 Gn: $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$
 Sol: $\vec{a} + \vec{b} = 4\hat{i} + 4\hat{j} + 0\hat{k}$
 $\vec{a} - \vec{b} = 2\hat{i} + 0\hat{j} + 0\hat{k}$
 $(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 4 & 0 \\ 2 & 0 & 0 \end{vmatrix} = \hat{i}(0) - \hat{j}(0) + \hat{k}(-8)$
 $= -8\hat{k}$
 Unit vector = $\frac{-8\hat{k}}{\sqrt{64}} = -\hat{k}$

Scalar triple product: (STP)

$$(\cdot, \times), \text{ Box product} = [\vec{a} \vec{b} \vec{c}]$$

The dot product of $\vec{a}$ and $\vec{b} \times \vec{c}$ is called STP written as $\vec{a} \cdot (\vec{b} \times \vec{c}) = [\vec{a} \vec{b} \vec{c}]$

$$* \vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a}, \vec{b}, \vec{c}]$$

$$* [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$* [\vec{a} \vec{b} \vec{c}] = -[\vec{a} \vec{c} \vec{b}] = -[\vec{b} \vec{a} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]$$

$$* [\vec{a} \vec{a} \vec{b}] = 0 = [\vec{a} \vec{b} \vec{b}] = [\vec{a} \vec{c} \vec{c}]$$

* Geometrically $[\vec{a} \vec{b} \vec{c}] = \text{Vol. of parallelepiped.}$

$$* \frac{1}{6} [\vec{a} \vec{b} \vec{c}] = \text{Vol. of tetrahedron}$$

* If $[\vec{a} \vec{b} \vec{c}] = 0$, 3 vectors are coplanar.

* If $[\vec{AB} \vec{AC} \vec{AD}] = 0$, gn. 4 points are coplanar.

EXERCISE 10.5

1] Find $[\vec{a} \vec{b} \vec{c}]$ if $\vec{a} = i - 2j + 3k$, $\vec{b} = 2i - 3j + k$, $\vec{c} = 3i + j - 2k$

$$\text{sol]} [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 1 & -2 & 3 \\ 2 & -3 & 1 \\ 3 & 1 & -2 \end{vmatrix} = 1(6-1) + 2(-4-3) + 3(2+9) \\ = 5 - 14 + 33 = 24$$

2] ST the vectors $\vec{a} = i - 2j + 3k$, $\vec{b} = -2i + 3j - 4k$, $\vec{c} = i - 3j + 5k$ are coplanar.

$$\text{sol]} \text{ Consider } [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 1(15-12) + 2(-10+9) + 3(6-3) \\ = 3 - 2 + 9 = 10 \neq 0$$

$$\therefore [\vec{a} \vec{b} \vec{c}] \neq 0$$

Gn vectors are coplanar.

** 3] Find λ if the vectors $\vec{i}-\vec{j}+\vec{k}$, $3\vec{i}+\vec{j}+2\vec{k}$ and $\vec{i}+\lambda\vec{j}-3\vec{k}$ are coplanar.

Sol] Gn: $\vec{a} = \vec{i}-\vec{j}+\vec{k}$, $\vec{b} = 3\vec{i}+\vec{j}+2\vec{k}$, $\vec{c} = \vec{i}+\lambda\vec{j}-3\vec{k}$

Gn: $\vec{a}, \vec{b}, \vec{c}$ are coplanar

$$\Rightarrow [\vec{a} \vec{b} \vec{c}] = 0$$

$$\begin{vmatrix} 1 & -1 & 1 \\ 3 & 1 & 2 \\ 1 & \lambda & -3 \end{vmatrix} = 0$$

$$1(-3-2\lambda) + 1(-11) + 1(3\lambda-1) = 0$$

$$-3-2\lambda-11+3\lambda-1 = 0$$

$$-15 + \lambda = 0$$

$$\boxed{\lambda = 15}$$

5] ST the four points with P.V.s $4\vec{i}+8\vec{j}+12\vec{k}$, $2\vec{i}+4\vec{j}+6\vec{k}$, $3\vec{i}+5\vec{j}+4\vec{k}$ and $\vec{i}+8\vec{j}+5\vec{k}$ are coplanar.

$$\text{Sol] } \vec{AB} = \vec{OB} - \vec{OA} = -2\vec{i} - 4\vec{j} - 6\vec{k}$$

$$\vec{AC} = \vec{OC} - \vec{OA} = -\vec{i} - 3\vec{j} - 8\vec{k}$$

$$\vec{AD} = \vec{OD} - \vec{OA} = \vec{i} + 0\vec{j} - 7\vec{k}$$

$$\text{Consider, } [\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} -2 & -4 & -6 \\ -1 & -3 & -8 \\ 1 & 0 & -7 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & -4 & -6 \\ -1 & -3 & -8 \\ 1 & 0 & -7 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 8 \\ 1 & 0 & -7 \end{vmatrix}$$

$$= 2 \{ 1(-21) - 2(-15) + 3(-3) \}$$

$$= 2 \{ -21 + 30 - 9 \}$$

$$= 2 \{ 0 \}$$

$$= 0$$

$$\therefore [\vec{AB}, \vec{AC}, \vec{AD}] = 0, \text{ gn 4 pts are coplanar.}$$

$$0 = 0$$

$$0 = 0$$

$$0 = 0$$

$$0 = 0$$

6) Find x such that the 4 points $A(3, 2, 1)$, $B(4, x, 5)$, $C(4, 2, -2)$ and $D(6, 5, -1)$ are coplanar.

Sol) $\vec{AB} = \vec{OB} - \vec{OA} = i + (x-2)j + 4k$

$\vec{AC} = \vec{OC} - \vec{OA} = i + 0j - 3k$

$\vec{AD} = \vec{OD} - \vec{OA} = 3i + 3j - 2k$

Given 4 pts are coplanar

$\Rightarrow [\vec{AB} \ \vec{AC} \ \vec{AD}] = 0$

$$\Rightarrow \begin{vmatrix} 1 & x-2 & 4 \\ 1 & 0 & -3 \\ 3 & 3 & -2 \end{vmatrix} = 0$$

$$\Rightarrow x = \frac{35}{7} = 5 = x$$

Eg 26] Find $\vec{a} \cdot (\vec{b} \times \vec{c})$ if $\vec{a} = 2i + j + 3k$, $\vec{b} = -i + 2j + k$, $\vec{c} = 3i + j + 2k$

Sol) $\vec{a} \cdot (\vec{b} \times \vec{c}) = [\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 2 & 1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 2 \end{vmatrix}$

$$= 2(4-1) - 1(-2-3) + 3(-1-6) = 6 + 5 - 21 = -10$$

Eg 28] Find λ if the vectors $\vec{a} = i + 3j + k$, $\vec{b} = 2i - j - k$, $\vec{c} = \lambda i + 7j + 3k$ are coplanar.

Sol) Given 3 vectors are coplanar

$\Rightarrow [\vec{a} \ \vec{b} \ \vec{c}] = 0$

$$\begin{vmatrix} 1 & 3 & 1 \\ 2 & -1 & -1 \\ \lambda & 7 & 3 \end{vmatrix} = 0$$

$$1(-3+7) - 3(6+\lambda) + 1(14+\lambda) = 0$$

$$4 - 18 - 3\lambda + 14 + \lambda = 0$$

$$\boxed{\lambda = 0}$$

Eg 29) ST the 4 pts A, B, C, D with PV's $4i + 5j + k$, $-(j + k)$, $3i + 9j + 4k$ and $4(-i + j + k)$ are coplanar.

Sol) Gn: $\vec{OA} = 4i + 5j + k$

$$\vec{OB} = -j - k$$

$$\vec{OC} = 3i + 9j + 4k$$

$$\vec{OD} = -4i + 4j + 4k$$

$$\vec{AB} = \vec{OB} - \vec{OA} = -4i - 6j - 2k$$

$$\vec{AC} = \vec{OC} - \vec{OA} = -i + 4j + 3k$$

$$\vec{AD} = \vec{OD} - \vec{OA} = -8i - j + 3k$$

$$\text{Consider } \begin{vmatrix} \vec{AB} & \vec{AC} & \vec{AD} \end{vmatrix} = \begin{vmatrix} -4 & -6 & -2 \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} = 0$$

$$= -4(12+3) - 6(-3+24) - 2(-3+32) = -4(15) - 6(21) - 2(29) = -60 - 126 - 58 = -244 \neq 0$$

$$\text{Eg 30) PT } (\vec{a} + \vec{b}) \cdot (\vec{b} + \vec{c}) \cdot (\vec{c} + \vec{d}) = 2[\vec{a} \vec{b} \vec{c}]$$

$$\text{LHS} = (\vec{a} + \vec{b}) \cdot (\vec{b} + \vec{c}) \cdot (\vec{c} + \vec{d})$$

$$= (\vec{a} + \vec{b}) \cdot \{(\vec{b} + \vec{c}) \times (\vec{c} + \vec{d})\}$$

$$= (\vec{a} + \vec{b}) \cdot \{(\vec{b} \times \vec{c}) + (\vec{b} \times \vec{d}) + (\vec{c} \times \vec{c}) + (\vec{c} \times \vec{d})\}$$

$$= \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{d}) + \vec{a} \cdot (\vec{c} \times \vec{d}) + \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{d}) + \vec{b} \cdot (\vec{c} \times \vec{d})$$

$$= [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{d}] + [\vec{a} \vec{c} \vec{d}] + [\vec{b} \vec{b} \vec{c}] + [\vec{b} \vec{b} \vec{d}] + [\vec{b} \vec{c} \vec{d}]$$

$$= [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{d}] + [\vec{a} \vec{c} \vec{d}] + [\vec{b} \vec{b} \vec{c}] + [\vec{b} \vec{b} \vec{d}] + [\vec{b} \vec{c} \vec{d}]$$

$$\text{Eg 30) PT } (\vec{a} + \vec{b}) \cdot (\vec{b} + \vec{c}) \cdot (\vec{c} + \vec{a}) = 2[\vec{a} \vec{b} \vec{c}]$$

$$\text{LHS} = (\vec{a} + \vec{b}) \cdot (\vec{b} + \vec{c}) \cdot (\vec{c} + \vec{a})$$

$$= (\vec{a} + \vec{b}) \cdot \{(\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})\}$$

$$= (\vec{a} + \vec{b}) \cdot \{(\vec{b} \times \vec{c}) + (\vec{b} \times \vec{a}) + (\vec{c} \times \vec{c}) + (\vec{c} \times \vec{a})\}$$

$$\begin{aligned}
 &= (\vec{a} + \vec{b}) \cdot \{(\vec{b} \times \vec{c}) + (\vec{b} \times \vec{a}) + (\vec{c} \times \vec{c}) + (\vec{c} \times \vec{a})\} \\
 &= \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{a}) + \vec{a} \cdot (\vec{c} \times \vec{a}) + \\
 &\quad \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) + \vec{b} \cdot (\vec{c} \times \vec{a}) \\
 &= [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{a}] + [\vec{a} \vec{c} \vec{a}] + [\vec{b} \vec{b} \vec{c}] + [\vec{b} \vec{b} \vec{a}] + [\vec{b} \vec{c} \vec{a}] \\
 &= [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{c}] \\
 &= 2[\vec{a} \vec{b} \vec{c}] \\
 &= \text{RHS}
 \end{aligned}$$

Eg 31] PT $[\vec{a} \vec{b} (\vec{c} + \vec{d})] = [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{d}]$

$$\begin{aligned}
 \text{LHS} &= [\vec{a} \vec{b} (\vec{c} + \vec{d})] \\
 &= \vec{a} \cdot (\vec{b} \times (\vec{c} + \vec{d})) \\
 &= \vec{a} \cdot \{(\vec{b} \times \vec{c}) + (\vec{b} \times \vec{d})\} \\
 &= \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{d}) \\
 &= [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{d}] \\
 &= \text{RHS}
 \end{aligned}$$

Eg 29] 3 vectors ~~are~~ $\vec{a}, \vec{b}, \vec{c}$ satisfy the condition $\vec{a} + \vec{b} + \vec{c} = 0$.
Evaluate the quantity $\mu = (\vec{a} \cdot \vec{b}) + (\vec{b} \cdot \vec{c}) + (\vec{c} \cdot \vec{a})$ if $|\vec{a}| = 1$,
 $|\vec{b}| = 4$ and $|\vec{c}| = 2$.

Sol] Gn: $\vec{a} + \vec{b} + \vec{c} = 0$

$$(\vec{a})^2 + (\vec{b})^2 + (\vec{c})^2 = 0$$

Squaring on both sides,

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$(1 + 4 + 2^2) + 2\mu = 0$$

$$21 + 2\mu = 0$$

$$\mu = \frac{-21}{2}$$

Eg 28] Let $\vec{a}, \vec{b}, \vec{c}$ be 3 vectors such that $|\vec{a}| = 3, |\vec{b}| = 4, |\vec{c}| = 5$ and each one of them being $\perp$ to the sum of the other two. Find $|\vec{a} + \vec{b} + \vec{c}|$.

Sol.] Qn: Each vector is $\perp$ to sum of the other two.

$$\begin{array}{l|l|l} \vec{a} \perp (\vec{b} + \vec{c}) & \vec{b} \perp (\vec{c} + \vec{a}) & \vec{c} \perp (\vec{a} + \vec{b}) \\ \vec{a} \cdot (\vec{b} + \vec{c}) = 0 & \vec{b} \cdot (\vec{c} + \vec{a}) = 0 & \vec{c} \cdot (\vec{a} + \vec{b}) = 0 \\ \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0 \rightarrow (1) & \vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{a} = 0 & \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} = 0 \end{array}$$

$$(1) + (2) + (3) \Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0 \rightarrow (4)$$

Consider $|\vec{a} + \vec{b} + \vec{c}|^2$

$$= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$\therefore |\vec{a} + \vec{b} + \vec{c}|^2 = 9 + 16 + 25 = 50$$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{50} = 5\sqrt{2}$$

*] PT the vectors $\vec{a} - \vec{b}$, $\vec{b} - \vec{c}$, $\vec{c} - \vec{a}$ are coplanar.

$$\begin{aligned} \text{Sol.] } [\vec{a} - \vec{b} \quad \vec{b} - \vec{c} \quad \vec{c} - \vec{a}] &= (\vec{a} - \vec{b}) \cdot [(\vec{b} - \vec{c}) \times (\vec{c} - \vec{a})] \\ &= (\vec{a} - \vec{b}) \cdot [(\vec{b} \times \vec{c}) - (\vec{b} \times \vec{a}) - 0 + (\vec{c} \times \vec{a})] \\ &= \vec{a}(\vec{b} \times \vec{c}) - \vec{a}(\vec{b} \times \vec{a}) + \vec{a}(\vec{c} \times \vec{a}) - \vec{b}(\vec{b} \times \vec{c}) + \vec{b}(\vec{b} \times \vec{a}) - \vec{b}(\vec{c} \times \vec{a}) \\ &= \vec{a}(\vec{b} \times \vec{c}) - \vec{b}(\vec{c} \times \vec{a}) \\ &= [\vec{a} \vec{b} \vec{c}] - [\vec{b} \vec{c} \vec{a}] \\ &= 0 \end{aligned}$$

$$* \quad |\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2$$